Geist in the Machine — Volume Spine

What Helps an AI Tutor Teach — and What Doesn't

Liam Magee[*]

Education Policy, Organization and Leadership
University of Illinois Urbana-Champaign

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Orientation

This project asks a straightforward question: can an AI tutor teach better if we change how it prepares an answer, rather than simply giving it a larger model?

We tried several ways of doing that. We gave the tutor an internal critic. We asked it to treat the learner as a thinking partner rather than an empty vessel. We gave it explicit records, pacing rules, strategy prompts, and — later — a small trained model responsible for one narrow teaching move.

The answers are mixed, which is exactly why the atlas exists. Some changes help. Some duplicate work the model already knows how to do. Some alter the tutor's behaviour without improving the lesson. This page tells the whole story at a human scale. Open a module when you want the evidence and technical detail.

The core idea

Give the tutor a second look. Two ideas started the programme.

The first is self-critique. Instead of sending the first answer it produces, the tutor drafts a reply, lets an internal critic ask what is wrong or missing, and then revises. We use Freud's

[*] This sentence is the only one actually authored by Liam Magee in this paper.

names — *ego* for the drafter and *superego* for the critic — but the everyday idea is simply "check your work before you speak."

The second is recognition. A tutor can treat a learner as a gap to be filled, or as a person whose reasoning is worth meeting — mistakes included. In practice that means asking for the learner's thinking and building from it, rather than writing over it. The philosophical source is Hegel, but one of our main questions was whether the philosophy words mattered at all.

From there the questions became concrete. Does the critic catch real mistakes? Does the learner-facing stance improve the answer or merely change the tone? Does a detailed model of the learner add anything the tutor cannot already infer? And when advice fails to change behaviour, can a small model be trained to own one move?

The findings

The findings fall into four broad families. Each label says how firmly we can hold the result. Here is the short version before the detail.

Meeting the learner raises the floor

[**SUPPORTED**] Supported by the current evidence

Asking the tutor to meet the learner's reasoning improves the weakest parts of its answer most and makes the answer more even. The effect remains when the Hegelian vocabulary is replaced with plain teaching language. The useful part is the stance toward the learner, not the theory words. The internal critic also catches and corrects real errors.

Caveat. The critic adds much more on weaker models than on stronger ones. With a capable model, the learner-facing stance already captures most of the gain.

Voice and drama can help — and crowd the learner out

[**EXPLORING**] Promising, but still exploratory

A parallel line gives the tutor a strong, performed voice and measures that presence separately from ordinary teaching quality. The design can produce a striking tutor voice. We are still testing whether productive internal tension can become a reliable teaching resource rather than just convincing role-play.

Caveat. Strong performance can crowd out the learner: the most charismatic tutors were not the best at drawing out the learner's own thinking. This line remains exploratory.

More machinery often changed behaviour, not outcomes

[**CLOSED**] Tested and closed under its stated gate

We gave the tutor increasingly detailed records of the learner, asked it to imagine alternative moves, prompted it to choose strategies, and routed its speaking stance. These additions often made the tutor act differently, but they did not reliably improve the lesson against a leaner baseline.

Caveat. One early advantage disappeared after a turn-counting error was fixed. The durable contribution is the testing method — especially the trap scenarios, explicit checks, and delivery audits — rather than a richer model of the learner.

A delivered move can reach the learner who resists

[**SCOPE-BOUND**] Supported within a narrow scope

A late line of studies built learners that resist — deferring every move, overclaiming, bored, or refusing the tutor's frame — and tested moves issued by machinery at detected moments. Delivery is the hinge: on the permission-seeking learner the same permission pasted into the standing prompt was never used once, while machinery issuing it at the moment changed the learner's conduct; on the overconfident learner the standing wording landed part-way, and the delivered gate still beat it. In the line's powered run, a delivered, state-matched move re-engaged almost every resistant learner.

Caveat. What the move wins depends on the shape of the resistance: bored learners went back to the work, while frame-refusers mostly stopped at naming a condition rather than acting on it. A fresh fifth depth calibration still failed reader agreement. Two successor calibrations did not answer whether any move converts that naming into doing: one finer endpoint failed reader agreement, and one satisfiable-demand persona never produced its registered trigger. A closing zero-call anchor rehearsal then located the repeated failure in the construct itself: re-anchored readers resolved the disputed votes but demoted the lineage's only unanimous rung-2 exemplars by the same logic, so the line closed — at this ladder's resolution the refuser's movement is graded concession, not a countable crossing. The naming-to-doing question is closed at this instrument, unmeasured beyond it, and all of it is simulated-learner evidence.

How we checked the work

The checking system matters as much as the ideas being tested.

The numbers check themselves. Important claims in the paper are tied to the stored evidence. If the paper's prose and data disagree, validation fails. This atlas is hand-written from that paper, so every module names its source sections and the empirical wording receives a separate claim audit before release.

We keep the working, not just the answer. Drafts, critiques, revisions, and delivery paths are recorded. That lets us ask why something helped, where it was lost, and whether the learner actually received the text that passed the check.

The ruler is part of the research. The tutoring rubric changed over four versions as we learned what "good tutoring" should mean in measurable terms. We treat that as part of the method, not as background maintenance.

Evidence map

Each module below opens one part of the paper. Filter by status, browse the short summaries, or open a PDF for the full evidence. The page and the PDFs come from the same map, so their labels stay together.

#	Module	Family	Status
1	Meeting the Learner Raises the Floor	Meeting the Learner	SUPPORTED
2	The Inner Critic Catches Real Mistakes	Meeting the Learner	SUPPORTED
3	The Tutor Did Not Become More Adaptive	Meeting the Learner	CLOSED
4	How We Checked Every Claim	How We Checked the Work	METHOD
5	Can Productive Tension Improve Teaching?	Voice, Drama, and Presence	PLANNED
6	A Strong Voice Can Crowd Out the Learner	Voice, Drama, and Presence	EXPLORING
7	A Bigger Learner Model Did Not Help	What the Tutor Needs to Know	CLOSED
8	The Ruler Is Part of the Research	How We Checked the Work	METHOD
9	The Human-Learning Question	What Comes Next	NEXT
10	The Model Still Sets the Ceiling	Meeting the Learner	SUPPORTED
11	Can a Teaching Move Travel?	Voice, Drama, and Presence	SCOPE-BOUND
12	More Layers Do Not Mean More Benefit	Meeting the Learner	SUPPORTED
13	Adaptive Speaking Style Did Not Hold Up	What the Tutor Needs to Know	SCOPE-BOUND
14	Advice, Enforcement, and the Cost of Control	What the Tutor Needs to Know	SCOPE-BOUND
15	The Hidden State Was Not Actually Hidden	How We Checked the Work	CLOSED
16	Training One Teaching Move into a Small Model	What Comes Next	SCOPE-BOUND
17	The Four Locks: Pricing Instrumentation, Casting the Seats, and Making Adaptation Measurable	What the Tutor Needs to Know	SCOPE-BOUND
18	Reaching the Learner Who Resists	Meeting the Learner	SCOPE-BOUND

#	Module	Family	Status
19	What Context Cannot Add, and What It Cannot Subtract	What the Tutor Needs to Know	SCOPE-BOUND

The modules

Module 1 — Meeting the Learner Raises the Floor

[**SUPPORTED**] Supported by the current evidence · *Meeting the Learner*

This module follows the programme's clearest prompt-level result: the learner-facing stance raises the floor rather than the ceiling. Matched teaching prompts show that the practical orientation, not the philosophical wording, carries the effect.

Sources: §3.2, §6.1, §7.9, §7.10

Module 2 — The Inner Critic Catches Real Mistakes

[**SUPPORTED**] Supported by the current evidence · *Meeting the Learner*

What the critic catches, how the tutor responds, and why two useful mechanisms do not simply add together. Across three models, their combined result falls about 15–17% short of simple addition.

Sources: §6.2, §6.4, §7.3

Module 3 — The Tutor Did Not Become More Adaptive

[**CLOSED**] Tested and closed under its stated gate · *Meeting the Learner*

A well-powered negative result: better prompts and self-critique improve answer quality, but did not make the tutor change course more effectively from turn to turn.

Sources: §6.3

Module 4 — How We Checked Every Claim

[**METHOD**] A contribution to how the research is done · *How We Checked the Work*

A tour of the audit trail: how critiques are classified, how important numbers are checked against stored evidence, how a result can be reproduced from source to claim, and why a study-wide lease and ledger are part of the empirical runtime.

Sources: §5.3, §5.9, §5.11, §7.4, §7.15

Module 5 — Can Productive Tension Improve Teaching?

[**PLANNED**] Planned work; evidence still to come · *Voice, Drama, and Presence*

An active line of work on disagreement, distinct roles, and resistance to easy agreement. The paper section is still being developed; this module remains a scaffold.

Sources: — (scaffold; prose pending)

Module 6 — A Strong Voice Can Crowd Out the Learner

[**EXPLORING**] Promising, but still exploratory · *Voice, Drama, and Presence*

A parallel family in which a durable source of voice repeatedly creates a fresh speaking persona. Charisma is measured separately from ordinary tutoring quality.

Sources: §6.7, §8.9

Module 7 — A Bigger Learner Model Did Not Help

[**CLOSED**] Tested and closed under its stated gate · *What the Tutor Needs to Know*

A mostly negative arc with a useful by-product: strong trap scenarios and an auditable runner. The tutor did not need a richer written portrait of the learner; the testing apparatus was the durable result.

Sources: §6.8, §6.9, §6.10

Module 8 — The Ruler Is Part of the Research

[**METHOD**] A contribution to how the research is done · *How We Checked the Work*

Four rubric versions, the problems they exposed, and why prompt tuning must be checked separately for each model family.

Sources: §5.2, §5.6, §7.7, §7.8

Module 9 — The Human-Learning Question

[**NEXT**] A future empirical question · *What Comes Next*

Why better-looking tutor responses are not the same as better human learning, and what the built-but-not-yet-recruited human pilot is meant to test.

Sources: §6.5, §8.1, §8.2

Module 10 — The Model Still Sets the Ceiling

[**SUPPORTED**] Supported by the current evidence · *Meeting the Learner*

Stronger base models leave less work for the critic. This module follows that pattern across the main model comparisons, the Writing Pad test, and the six-model check that did not find the predicted helpful-then-harmful profiling pattern.

Sources: §5.7, §5.8, §6.6, §8.3

Module 11 — Can a Teaching Move Travel?

[**SCOPE-BOUND**] Supported within a narrow scope · *Voice, Drama, and Presence*

A18 remains the main evidence: 10 of 14 bounded cases. A19 found two promising n=1 candidates in one model family, but neither repeated in a two-seed stability check. There is no pooled A19 rate or confirmed transfer effect, and no human-learning, deployment, or weight-learning claim.

Sources: §7.9

Module 12 — More Layers Do Not Mean More Benefit

[**SUPPORTED**] Supported by the current evidence · *Meeting the Learner*

A pre-registered comparison of a small kernel and the full mechanism stack across resistance, suggestion, and trap suites, at n=6 per cell-scenario. It explains why several good instructions often substitute for one another instead of adding up.

Sources: §6.14, §7.11

Module 13 — Adaptive Speaking Style Did Not Hold Up

[**SCOPE-BOUND**] Supported within a narrow scope · *What the Tutor Needs to Know*

A 120-dialogue confirmatory study in one simulated detective world. It closes this field selector on the present evidence, while keeping the result separate from any claim about an isolated tutor model or human learning.

Sources: §6.17, §6.29

Module 14 — Advice, Enforcement, and the Cost of Control

[**SCOPE-BOUND**] Supported within a narrow scope · *What the Tutor Needs to Know*

A ladder from "remember this advice" to "do this now" to a compiled constraint. Better-timed information still missed its behavioural bar. Enforcement changed the action by taking the choice away from the model, but that result is descriptive and paid for its control elsewhere.

Sources: §6.15, §6.16, §6.18

Module 15 — The Hidden State Was Not Actually Hidden

[**CLOSED**] Tested and closed under its stated gate · *How We Checked the Work*

A zero-call reanalysis showed that the first negative result was partly a data-starved instrument, then exposed a deeper problem: the synthetic benchmark was fully observable. A future sensor test needs a genuinely ambiguous substrate.

Sources: §6.19

Module 16 — Training One Teaching Move into a Small Model

[**SCOPE-BOUND**] Supported within a narrow scope · *What Comes Next*

The trained model nearly learned the missing warrant cue in its weights. In the final live run, the fully checked committee reached 0.477 (42/88) against 0.136 (22/162) for pooled controls, +0.341 with CI95 [0.223, 0.457], at no coverage cost and seam parity. The result is development-tier, one move, two worlds, one live tutor family, simulated learners, and no human-learning or deployment claim.

Sources: §6.20, §6.21, §6.22

Module 17 — The Four Locks: Pricing Instrumentation, Casting the Seats, and Making Adaptation Measurable

[**SCOPE-BOUND**] Supported within a narrow scope · *What the Tutor Needs to Know*

One clause of information placement moved bare-tutor closure more than any tutoring behaviour observed (5/5 to 4/5 on a minimal-pair world). Planted, typed learner states with human-adjudicated repairs made repair delivery a measured quantity; a deterministic pressure trigger graduated from 6/20 to 17/20 against that gold in one tuning pass. The move-card mechanism delivered 63% right-repairs on covered moments against the baseline's 42%, with closures and leak discipline intact — k=3, one world, one persona, simulated learner, no human-learning claim. The disclosure result is the instrument finding: turn-local rubrics under-measure adaptation by construction, and one added sentence repairs half the deficit.

Sources: §6.23, §6.24

Module 18 — Reaching the Learner Who Resists

[**SCOPE-BOUND**] Supported within a narrow scope · *Meeting the Learner*

Four studies walk one idea across opposite deficit shapes. On a permission-seeking learner, the gate's always-on per-turn steering carried the deference-break effect while the sensor-timed challenge paid separately in decision correctness (about twelve points), and the gate's wording pasted as a standing instruction delivered zero challenges in twenty-four dialogues. On an overconfident learner a later, sealed re-analysis of the readers' own judgments — a post-hoc label the finding carries everywhere — located the effect in epistemic state, not voice: warranted commitment shifts more than doubled under the gate while both voice-pinned conduct channels stayed flat, and the registered primary endpoint closed late, directional only, under a disclosed instrument amendment. Five registered studies on a bored learner returned no move claim — a window that could not be reached, then a contrast that was not delivered — and closed as a measurement result. The powered two-face run then held delivery to exhaustive checks and found re-engagement near ceiling on both faces, with the boredom face doing the work and the refusal face naming its terms. A fresh fifth depth calibration repaired the output contract but still failed endpoint agreement, so its one treatment rung-2 case is descriptive rather than a treatment result. A separate 24-dialogue reader calibration failed agreement twice, so its conflicting arm spreads were not pooled; a separate 48-dialogue calibration produced zero determinate endpoint rows because the intended learner challenge never appeared. A closing zero-call anchor rehearsal located those repeated agreement failures in the construct itself and closed the depth line: the refuser's genuine movement is graded concession below the binary ladder's resolution, and the powered run's descriptive depth split stands as recorded. These are instrument and persona-trigger failures, not evidence of an effect or a null. Development-tier throughout: simulated learners, one model family in the speaking seats, no human-learning claim.

Sources: §6.25, §6.26, §6.27, §6.28, §6.30

Module 19 — What Context Cannot Add, and What It Cannot Subtract

[**SCOPE-BOUND**] Supported within a narrow scope · *What the Tutor Needs to Know*

A discussion-tier synthesis, not a new measurement. Prompt-level adaptation selects among completions the model already affords; it does not construct new capacities and it does not delete trained ones. The practical consequence for designers: an arm that needs the model to be less forthcoming than its training should be verified for delivered conduct before scoring, or declared unavailable. External work on preference training, negated instructions, and persona drift places the boundary where these studies found it.

Sources: §7.16

How to read the status labels

- [**SUPPORTED**] — Supported by the current evidence
- [**SCOPE-BOUND**] — Supported within a narrow scope
- [**EXPLORING**] — Promising, but still exploratory
- [**CLOSED**] — Tested and closed under its stated gate
- [**PROPOSITION**] — A theoretical proposal, not a result
- [**METHOD**] — A contribution to how the research is done
- [**PLANNED**] — Planned work; evidence still to come
- [**NEXT**] — A future empirical question

A short glossary

Ego / superego. Our names for the drafter and the internal critic. The drafter keeps final say; the critic checks the draft before it reaches the learner.

Recognition. Treating the learner as a thinking partner, not a gap to fill: ask for their reasoning and build on it.

Calibration. Raising the weakest parts of an answer and making its quality more even — lifting the floor more than the ceiling.

Claim-status. A short label saying how firmly we can hold a result and where its limits begin.

Trap suite. A set of lessons designed to tempt the tutor into a known mistake, so we can see whether it notices and changes course.

Provable discourse. Automatic checks that keep the paper's important numbers tied to the evidence behind them.

Id-director. A design that gives the tutor a durable source of voice and repeatedly turns it into a fresh speaking persona.

What's still open

Two limits matter as much as the results.

First, these studies mostly judge tutor responses with language models while the tutor speaks to simulated learners. We do not yet know whether the supported changes improve learning for real people. The human pilot is built, but recruitment waits on approval, final consent language, and final learning materials.

Second, a strong base model still sets the ceiling. An internal critic can add a lot to a weaker model and almost nothing to a stronger one. The small-model training programme produced partial traction, and the live committee passed its scoped delivery test only after every text path it owned was checked; the hard-safety guardrail still failed. These are bounded research results, not deployment claims.